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Class: 3rd Year
Department: Cyber Security

Unit 4 – Cloud Security Assignment

Topic: Launch a Windows VM and Connect Using RDP

1. Aim

To launch a Windows Server virtual machine on AWS EC2, restrict Remote Desktop Protocol (RDP, port 3389) to my own IP address, connect to the server over RDP, explore the system, create a new user with a strong password, enable and verify Windows Firewall, create a file on the server, and document the security configuration.

2. Architecture Overview

Component	Choice
Cloud provider / service	AWS / Amazon EC2
Operating system	Microsoft Windows Server 2022 Base
Instance type	t2.micro or t3.micro (Free Tier eligible for new accounts)
Remote access	RDP (TCP 3389), restricted to my public IP (/32)
Authentication	Key pair used to decrypt the Administrator password, then a separate low-privilege user

Traffic flow: My PC (RDP client) → Internet → Security Group (allows 3389 from my IP only) → Windows Firewall → Windows Server.

3. Step 1 – Launch a Windows Server Instance

1. Open the **AWS Management Console → EC2 → Launch instance**.
2. Configure:

Setting	Value
Name	`windows-rdp-server`
AMI	Microsoft Windows Server 2022 Base
Instance type	`t2.micro` (or `t3.micro`)
Key pair	Create new → `win-rdp-key`, type RSA, format `.pem`
Storage	30 GiB gp3 (default for Windows)

3. Keep the `.pem` file safe. It is needed to retrieve the Administrator password.

[Insert Screenshot 1: Launch instance page showing Windows Server 2022 AMI and instance type]

Figure 1: Selecting the Windows Server AMI and instance type

4. Step 2 – Configure the Security Group (RDP Only From My IP)

Under **Network settings → Edit**, create a security group `rdp-sg` with exactly one inbound rule:

Type	Protocol	Port	Source	Purpose
RDP	TCP	3389	**My IP** (`x.x.x.x/32`)	Remote administration from my machine only

Security reasoning:

- RDP must **never** be exposed to `0.0.0.0/0`. Open RDP ports are constantly scanned and brute-forced, and are a common ransomware entry point.
- Selecting **My IP** restricts the rule to a single `/32` address, so only my connection can reach port 3389.
- No other inbound ports are opened (principle of least privilege). Security Groups deny everything not explicitly allowed.
- If my home or college IP changes, the rule must be updated; it should not be widened.

[Insert Screenshot 2: Security Group inbound rule: RDP 3389, source = My IP /32]

Figure 2: RDP restricted to a single trusted IP address

Launch the instance and wait for **Instance state = Running** and **2/2 status checks passed**.

[Insert Screenshot 3: Instances list showing Running state and public IPv4 address]

Figure 3: Windows instance running

5. Step 3 – Connect Using Remote Desktop Protocol (RDP)

Get the Administrator password:

1. Select the instance → **Connect** → **RDP client** tab.
2. Click **Get password** and upload the `.pem` key file → **Decrypt password**. (Windows instances can take about 4 minutes after launch before the password is available.)
3. Note the public DNS/IP, the username `Administrator`, and the password. The password is **not** included in this report.

Connect:

- On Windows: open **Remote Desktop Connection** (`mstsc`), enter the public IPv4 address, click **Connect**, and sign in as `Administrator`. Accept the certificate warning.
- Alternatively, download the `.rdp` shortcut file from the console and open it.

Connection details (passwords excluded)

Item	Value
Protocol / port	RDP / TCP 3389
Public IPv4 address	*(fill in from your instance)*
Username (initial)	Administrator
Password	*Not disclosed*
Region	*(fill in, e.g. ap-south-1)*

[Insert Screenshot 4: Connect → RDP client tab (Get password / decrypt dialog). Hide the password.]

Figure 4: Retrieving the connection details (password hidden)

[Insert Screenshot 5: Remote Desktop Connection window with the instance's IP entered]

Figure 5: Connecting with the Remote Desktop client

6. Step 4 – Explore the Desktop and Check System Information

After login, **Server Manager** opens automatically. Explore the desktop, Start menu, and Server Manager dashboard. Then open **PowerShell** (Start → right-click → *Windows PowerShell* → Run as administrator) and run:

```
hostname
whoami
systeminfo | Select-String "OS Name","OS Version","System Type","Total Physical Memory"
Get-ComputerInfo | Select-Object WindowsProductName, OsVersion, CsProcessors
ipconfig
```

You can also view **Settings → System → About**, or run `winver`.

[Insert Screenshot 6: Server Manager dashboard / Windows Server desktop]

Figure 6: Windows Server desktop after RDP login

[Insert Screenshot 7: PowerShell showing systeminfo / hostname / ipconfig output]

Figure 7: System information gathered from the server

7. Step 5 – Create a New User Account with a Strong Password

Using the built-in Administrator for everyday work is unsafe. A separate standard account is created and allowed to use Remote Desktop:

```
# Prompt for the password securely (nothing is stored in the command history)
$pw = Read-Host -AsSecureString "Enter a strong password for labuser"

# Create the user
New-LocalUser -Name "labuser" -Password $pw -FullName "Lab User" `
  -Description "Standard user for Unit 4 assignment" -PasswordNeverExpires:$false

# Allow RDP login, but do NOT make this user an administrator
Add-LocalGroupMember -Group "Remote Desktop Users" -Member "labuser"

# Verify
Get-LocalUser -Name labuser
Get-LocalGroupMember -Group "Remote Desktop Users"
```

Strong password requirements used: at least 14 characters, a mix of upper-case, lower-case, digits and symbols, no dictionary words or personal details, and unique to this server. Windows Server also enforces password complexity by default.

Account lockout policy (protects against password guessing):

```
net accounts /lockoutthreshold:5 /lockoutduration:30 /lockoutwindow:30
net accounts
```

[Insert Screenshot 8: PowerShell showing the new user created (Get-LocalUser)]

Figure 8: New standard user account created (password not shown)

8. Step 6 – Enable Windows Firewall and Verify Its Status

```
        # Check current status of all three profiles
        Get-NetFirewallProfile | Select-Object Name, Enabled

        # Enable the firewall on all profiles
        Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True

        # Verify again
        Get-NetFirewallProfile | Select-Object Name, Enabled, DefaultInboundAction

        # Confirm Remote Desktop rules exist and are enabled (so RDP keeps working)
        Get-NetFirewallRule -DisplayGroup "Remote Desktop" |
          Select-Object DisplayName, Enabled, Direction, Action
```

Expected result: `Enabled : True` for Domain, Private and Public, with default inbound action *Block*. The firewall can also be checked graphically via **Windows Defender Firewall with Advanced Security** (`wf.msc`).

> **Important:** Never disable the firewall. The Security Group protects the network edge; Windows Firewall protects the host itself (defense in depth).

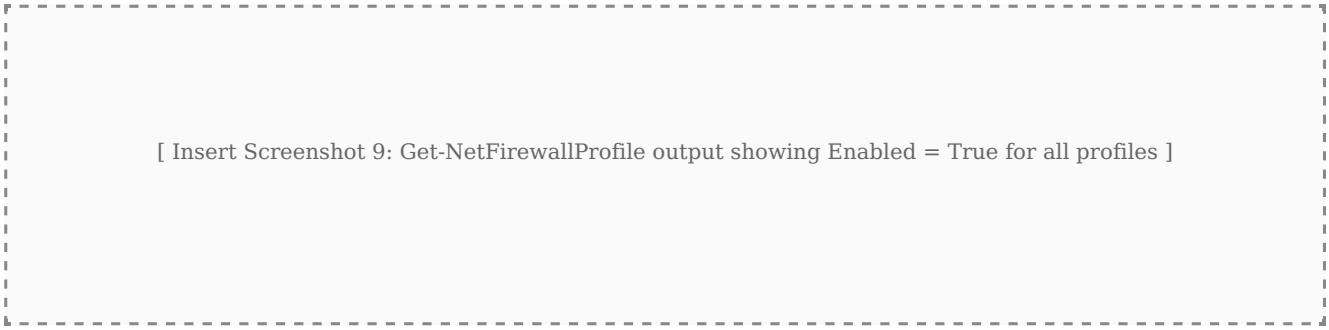


Figure 9: Windows Firewall enabled and verified

9. Step 7 – Create a Text File on the Server

```
        New-Item -Path "C:\Lab" -ItemType Directory -Force
        Set-Content -Path "C:\Lab\hello.txt" -Value "Unit 4 - Cloud Security Assignment. Windows VM on AWS EC2
accessed via RDP."
        Get-Content "C:\Lab\hello.txt"
```

The same can be done by opening **Notepad**, typing a message, and saving it as `C:\Lab\hello.txt`.



Figure 10: Text file created on the Windows server

10. Security Configurations Summary

#	Control	Implementation	Benefit
1	Network restriction	RDP 3389 allowed only from My IP (/32)	Prevents internet-wide scanning and brute force
2	No other inbound ports	Security Group has a single rule	Minimal attack surface
3	Key-protected admin password	Administrator password retrievable only with the `.pem` key	Credentials are not guessable or shared
4	Separate low-privilege user	`labuser` in *Remote Desktop Users* only	Daily work without admin rights
5	Strong password + lockout	14+ chars, complexity, lockout after 5 failures	Defends against password guessing
6	Host firewall	Windows Firewall enabled on all profiles	Defense in depth beyond the Security Group
7	Passwords not documented	No credentials appear in this report	Prevents credential leakage

Further recommendations: keep Windows Update enabled, keep Network Level Authentication (NLA) on for RDP, use an Elastic IP or AWS Systems Manager Session Manager / a VPN or bastion host for production access, and enable MFA for the AWS account.

11. Troubleshooting

Problem	Likely cause	Fix
RDP connection times out	My public IP changed, or no 3389 rule	Edit Security Group and set RDP source to **My IP** again
“Get password” says not available	Instance still initialising	Wait about 4 minutes after launch and retry
Credentials rejected	Wrong user or password	Use `Administrator` with the decrypted password; re-decrypt with the correct `.pem`
New user cannot RDP in	Not in the right group	`Add-LocalGroupMember -Group "Remote Desktop Users" -Member labuser`

12. Clean-up (to Avoid Charges)

After the assignment is evaluated, go to **EC2 → Instances → select the instance → Instance state → Terminate instance** (or **Stop** if the instance is still needed). Release any Elastic IP and delete unused volumes and security groups. Windows instances cost more than Linux ones, so do not leave them running.

13. Conclusion

A Windows Server 2022 instance was launched on AWS EC2 and secured with a Security Group that allows RDP only from a single trusted IP address. The server was accessed through RDP, system information was reviewed, a separate low-privilege user with a strong password and account lockout policy was created, Windows Firewall was enabled and verified, and a text file was created on the server. The exercise demonstrates core cloud security practices: restricting remote-access exposure, least privilege, strong authentication, defense in depth, and not exposing RDP to the open internet.